

Ayna Musayeva

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PROFILE

An investigative researcher specializing in speculative markets, concealed ownership structures, complex financing arrangements, and opaque financial operations within the global art market, with particular expertise in regulatory frameworks, legislation, and cross-border compliance.

Founder and AI Research Infrastructure Developer of Nimillux, an independently designed and developed, fully local AI research and intelligence system built around verification, provenance, and source-grounded analysis.

SELECTED RESEARCH & TECHNICAL PROJECTS

Capital, Ownership, and Opacity: Investigative & Regulatory Research Initiative (2024 – Present)

A comprehensive investigative and regulatory research initiative examining ownership, financing, and systemic opacity across the global art market

- Performed clause-level statutory and regulatory analysis across international, federal, and art-market-specific legal frameworks, identifying and annotating the precise legislative wording that leaves illicit financial activity unaddressed
- Led a comprehensive legal and historical analysis of international art-restitution law, revealing a systemic loophole, recurring across every major treaty and legislative instrument, that lets ownership vanish into opacity through an ordinary sale rather than theft
- Conducted a rigorous, independent audit of widely cited market statistics
- Identified and disclosed a material, unresolved sourcing conflict within a central historical account
- Conducted a formal regulatory-classification analysis of art investment funds spanning multiple legal frameworks: U.S. federal securities law under the SEC v. Howey Co. test, art-fund-specific regulatory regimes, and UNESCO cultural-property conventions
- Delivered a wide-ranging, cross-jurisdictional analysis of fund structuring, registration, and tax implications across major offshore jurisdictions
- Exposed a systemic disclosure and oversight deficiency, rooted in permissively worded regulation, that leaves art investment funds outside the oversight applied to hedge funds despite structurally comparable architecture
- Uncovered undisclosed family ties connecting senior auction-house executives to political leadership, a previously unreported finding

Nimillux: AI Platform (2024 – Present)

A production-grade, fully local AI research platform with built-in hallucination detection and citation tracking across large-scale document corpora

- Independently sourced, vetted, and curated a research corpus of tens of thousands of documents spanning a sprawling range of technical and financial disciplines, applying the same primary-source research discipline as the investigative work above, conducted entirely without external or AI assistance
- Directed the architecture of a fully local AI infrastructure with zero external API dependency, eliminating third-party data exposure and provider-side content-filtering risk
- Architected a hybrid retrieval layer combining dense semantic search with lexical matching and domain-aware routing, engineered to prevent any single subject area from dominating results across a large, heterogeneous multi-domain corpus
- Engineered a dual-layer hallucination-detection system, independently verifying each generated claim against its source and building a second, architecturally distinct verification model to close shared blind spots between retrieval and verification
- Identified a critical blind spot in the system's binary pass/fail verification model and drove the redesign into a tiered confidence framework, after an audit revealed it was masking weakly-supported claims inside answers reported as fully verified
- Directed resolution of a broad class of query-understanding failures, spanning negation, imperative phrasing, symbolic notation, and typographical variance, that were causing valid queries to silently fail
- Built and maintained a hand-labeled regression-testing suite that grew continuously alongside the platform, catching and rejecting a flawed pattern-matching implementation before it reached production
- Designed a pre-generation trust-gating layer that abstains from responding entirely when retrieval confidence is insufficient, rather than allow the model to hallucinate around irrelevant context

- Built a dedicated citation-validation layer that catches fabricated reference identifiers invisible to topic-level semantic grounding
- Established the platform's hardware operating envelope through direct empirical stress-testing of model capacity on consumer-grade infrastructure, rejecting assumed limits in favor of measured failure thresholds
- Defined requirements for a full-featured conversational interface with persistent multi-turn memory, cross-document synthesis, and a dedicated contradiction-detection mode

PROFESSIONAL EXPERIENCE

Copywriter, Apogee Media (2023 – 2024)

- Produced content across publications, video scripts, and web platforms, partnering with marketing and design teams to maintain consistent brand voice across campaigns
- Adapted messaging and tone across audience segments and content formats, revising copy against stakeholder feedback and brand style guidelines under campaign deadlines

Editor, Musée Magazine (2018 – 2019)

- Curated and edited the emerging-artist roster for the magazine's 21st issue, interviewing photojournalists and artists for weekly publication
- Managed freelance contributor relationships and the editorial production schedule
- Edited and fact-checked contributor submissions for publication, coordinating with design and photo teams to finalize layout under weekly deadlines

Gallery Assistant, Frieze Art Fair (2019)

- Handled artwork installation, client-facing sales, and logistics for exhibiting galleries
- Maintained collector and artist contact records to support post-fair follow-up
- Coordinated with exhibiting galleries on artwork presentation, condition reporting, and insurance documentation to support fair-day logistics

EDUCATION

MFA, New York Academy of Art (2018)

BA, Western Caspian University (2016)

CERTIFICATES

Documentary Practices and Visual Journalism, International Center of Photography (2022)

Creative Practices, International Center of Photography (2020)

ADDITIONAL SKILLS

Research & Editorial: primary-source verification, open-source intelligence (OSINT) research, financial and regulatory analysis (including bookkeeping and financial-statement review), academic citation, publication management

Technical: Python, retrieval-augmented generation (RAG) architecture, vector search (FAISS), OCR and document processing, local/offline LLM deployment

LANGUAGES

English (professional working proficiency) • Russian (fluent) • Turkish (fluent) • Azerbaijani (fluent)